

NON-PERMANENCE RISK REPORT DARKWOODS FOREST CARBON PROJECT

The Nature Conservancy of Canada

Document Prepared By: 3GreenTree Ecosystem Services Ltd.

Project Title	Darkwoods Forest Carbon Project
Version	v.2
Date of Issue	01-Nov-2017
Project ID	607
Monitoring Period	01-Jan-2013 to 31-Dec-2016
Prepared By	3GreenTree Ecosystem Services Ltd. For Nature Conservancy of Canada
Contact	Mike Vitt, 24-3871 River Rd West, Delta, BC, Canada V4K 3N2. +1 780-998-5478. mike.vitt@3greentree.com , www.3greentree.com .

Table of Contents

1	Internal Risk	3
1.1	Project Management	3
1.2	Financial Viability	3
1.3	Opportunity Cost	3
1.4	Project Longevity	4
2	External Risk	4
2.1	Land Tenure and Resource Access Impacts	4
2.2	Community Engagement	4
2.3	Political Risk	4
3	Natural Risk	7
3.1	Natural Risk	7
3.1.1	Fire Significance	7
3.1.2	Fire Likelihood	7
3.1.3	Fire Mitigation	7
3.1.4	Pest and Disease Significance	7
3.1.5	Pest and Disease Likelihood	8
3.1.6	Pest and Disease Mitigation	8
3.1.7	Extreme Weather Significance	9
3.1.8	Extreme Weather Likelihood	9
3.1.9	Extreme Weather Mitigation	9
3.1.10	Other Natural Risks	10

1

INTERNAL RISK

1.1 Project Management

- a) Darkwoods utilizes 100% native species in reforestation using locally sourced seed stock and following typical forest regeneration practices in British Columbia. Multiple species are planted in a manner consistent with local forest types.
- b) The project area has controlled access with locked gates on road access points. British Columbia has excellent private property rights and enforcement agencies. All timber harvesting requires unique timber marks on each load, and timber theft is rare. No specific ongoing enforcement is required to protect carbon stocks on Darkwoods from illegal or unplanned activities.
- c) NCC has extensive experience with forest conservation and property management, and directly employs foresters, biologists, and other conservation professionals and technicians on an ongoing basis. The project team, as listed in the Monitoring Report, Section 2.4.1, the project team contains individuals with significant experience in all areas related to project activities. All listed team members have well over 5 years experience in the fields of forestry, biology, finance, conservation, etc., including specific experience managing the Darkwoods project for over 5 years. For example, Mike Vitt has over 23 years of forestry and forest management experience, and over 8 years in carbon project development and management. Other team members exceed this level of experience in biology, forestry, and conservation.
- d) NCC staffs an office within a 30 minute drive of the property in Nelson, BC, and maintains other offices and staff in multiple locations in British Columbia, including regional headquarters in Victoria, BC (short flight away).
- e) NCC management staff has over 8 years experience with the carbon project on Darkwoods. Implementation partner 3GreenTree has more than 7 years experience with VCS project design, development and management, including successful validation and verification on multiple VCS projects, development and management of a VCS AFOLU methodology (VM0012), and project design under other carbon certification systems.
- f) NCC property management planning is a detailed and active program of adaptive management that is updated on a regular basis to incorporate monitoring and other new information and data over time. As described in the joint VCS/CCB Monitoring Report, NCC continues to monitor, research, and update management plans and project results for Darkwoods, including (for example) recent ongoing work related to key species habitat protection and management (i.e. caribou, grizzly, white bark pine, etc.). The adaptive management plans relevant to this monitoring period are the document: Darkwoods Conservation Area Property Management Plan 2011-2016 (February 2011); which has been updated to the current document: Darkwoods Conservation Area Property Management Plan, British Columbia Region 2016 – 2020 (May 2016).

1.2 Financial Viability

- d) The Darkwoods project remains well above break-even for this monitoring period (2013-2016), as shown in the file: "Item 4(a) re Break Even Analysis - Carbon Project Profitability - Final For SCS Audit - September 20, 2017.xls". As of 2016, the project is >\$4.5 million above an estimate of break-even. NCC has established and funded property endowment funds from the proceeds of project sales, which will continue to fund the project expenses on an ongoing basis. Therefore the project cash flow breakeven point is 4 years or less from the current risk assessment (d).
- h) As above, the project is strongly cashflow positive from current sales, and already well above cash flow break even. Therefore the project has be default secured 80% or more of the funding needed to cover the total cash out before reaching breakeven (h).

1.3 Opportunity Cost

- d) The estimated NPV of the project scenario is shown to be at least 50% more profitable than the baseline scenario NPV, as shown in the file “NCC Financial Model - NPV Valuation 2017 update v2”. This scenario shows a baseline logging scenario IRR of 4.7%, which is slightly below typical timberland returns (5.25-6.75%), it is a reflection of the somewhat higher acquisition cost for the property, and it results in a negative NPV at 5.25% cost of capital (as expected). The carbon scenario is NPV positive and more profitable, mainly due to lower operating costs/higher margins in a carbon project. The carbon price scenario assumed is relatively conservative, comparable to current AFOLU voluntary prices and well below CA ARB market projections; however, in reality the voluntary carbon market is fairly uncertain, and therefore to be conservative, the project has assessed the Opportunity Cost risk at the more neutral d) (instead of the calculated f) rating) for a score of 0. This assumes the baseline and project would be +/-20% of each other in NPV returns.
- g) NCC is a non-profit organization
- h) As part of implementing the project scenario, NCC and the previous landowner registered the Darkwoods project in the Canadian Ecological Gifts Program, which provides legally binding commitments to retain the property under conservation management in perpetuity, in full support of the project activities.
- i) The Canadian Ecological Gifts Program protects the project activities in perpetuity (i.e. >100 years).

1.4 Project Longevity

- b) The Darkwoods project has a 100-year duration, and the Canadian Ecological Gifts Program is a legally binding agreement to continue conservation management on the property in perpetuity.

2 EXTERNAL RISK

2.1 Land Tenure and Resource Access Impacts

- a) The entire Darkwoods project area is owned in fee simple by NCC.
- c) There are no title disputes or other ownership disputes on Darkwoods.
- d) NCC holds various mutual road use agreements with adjacent landowners. No access or use rights disputes exist.
- e) n/a
- f) Darkwoods participates in the Canadian Ecological Gifts Program which provides a legal commitment to manage the property for conservation (and therefore protect carbon stocks) in perpetuity.
- g) n/a

2.2 Community Engagement

- a) There are no communities living within the project area.
- b) There are no households within 20km who are reliant on the project area. NCC has an ongoing program of periodic engagement with interested stakeholders in the project zone, and they participate in many regional community initiatives and public processes related to conservation, land use planning, and the like.
- c) As per the CCB PDD, the Darkwoods project provides a net positive impact to the social and economic well-being of local communities in the surrounding project zone area. However, as there are no community members dependent on the Darkwoods property for their livelihoods, then Darkwoods has not claimed the mitigation credit.

2.3 Political Risk

- e) As per Table 1, the Governance score of 1.63 is >0.82.
- f) The mitigation is not applicable.

Table 1 - World Bank Governance Indicators for Canada 2011-2015 (Source: <http://info.worldbank.org/governance/>).

Governance Indicator	Year	Governance Score	5 Year Avg.
Voice and Accountability	2011	1.41	1.44
	2012	1.45	
	2013	1.47	
	2014	1.43	
	2015	1.44	
Political Stability/Absence of Violence	2011	1.06	1.12
	2012	1.09	
	2013	1.03	
	2014	1.16	
	2015	1.24	
Government Effectiveness	2011	1.78	1.77
	2012	1.77	
	2013	1.78	
	2014	1.76	
	2015	1.77	
Regulatory Quality	2011	1.69	1.73
	2012	1.7	
	2013	1.73	
	2014	1.83	
	2015	1.71	
Rule of Law	2011	1.74	1.80
	2012	1.76	
	2013	1.75	
	2014	1.89	
	2015	1.84	
Control of Corruption	2011	2	1.90
	2012	1.94	
	2013	1.89	

	2014	1.82	
	2015	1.85	
Overall Average Score:			1.63

3 NATURAL RISK

3.1 Natural Risk

a) Fire:

3.1.1 Fire Significance

Darkwoods is located in a region of B.C. known as the “Interior Wet Belt” and the property contains a highly diverse set of micro- and macro-site conditions that strongly reduce catastrophic fire risk. The fire return interval for the ICH zone (Interior Cedar-Hemlock) is 150-200 years, and the ESSF zone (Englemann Spruce-Subalpine Fir) is 150-350 years (B.C. Ministry of Forests, 2003). Devastating fires are rare, with forest diversity and complexity limiting campaign type fires (Wong, 2003). The Darkwoods property has a history of successful fire protection with <600ha of total fire disturbance recorded over the past 40 years. The 5 fire occurrences in the current monitoring period impacted 182.8ha, (0.3% of the project area), and only a portion of the carbon in these burned forests was released. The rating of insignificant remains appropriate.

Rating: Insignificant.

3.1.2 Fire Likelihood

Darkwoods does encounter incidents of fire on an annual or semi-annual basis, primarily from occasional lightning strikes. However, Darkwoods is located in a region of B.C. known as the “Interior Wet Belt” and the property contains a highly diverse set of micro- and macro-site conditions that strongly reduce catastrophic fire risk. The fire return interval for the ICH zone (Interior Cedar-Hemlock) is 150-200 years, and the ESSF zone (Englemann Spruce-Subalpine Fir) is 150-350 years (B.C. Ministry of Forests, 2003).

Darkwoods has had 5 recorded lightning strike fires since the project start in 2008. In 2014 and 2015 a total of 5 fires burned a total of 182.8ha. This supports the risk rating that fires occur less than every 10 years.

Rating: < Every 10 years

3.1.3 Fire Mitigation

The Darkwoods property has a history of successful fire protection with <600ha of total fire disturbance recorded over the past 40 years. The property is fully roaded with all weather access roads, and property access is strictly controlled. NCC has undertaken significant fire risk mitigation steps by entering into a fire protection agreement with the B.C. government (who have an initial attack heli-base stationed in nearby Creston, B.C.). NCC employs forestry staff and stores equipment and resources at the nearby Darkwoods office in Nelson, B.C. Ongoing forestry activities on Darkwoods are utilized as part of the property management plan, with harvesting activities being used to reduce fire hazard, provide access and fire breaks, reduce or salvage high mortality stands, and/or to continue to provide long term stand and age diversity across the property. During extreme fire conditions, Darkwoods typically follows provincial guidance and shuts down all operations and restricts any access onto the property until conditions improve.

The forest fire detection, initial attack, and fire containment all worked appropriately and as planned, demonstrating the partnership with nearby B.C. Fire Fighting assets is effective.

Rating: Both of the Above

b) Pests and Disease Outbreaks:

3.1.4 Pest and Disease Significance

Darkwoods is located in a transition zone between wet and dry climates, and has a high level of elevation, terrain, and stand and tree diversity. Therefore, the property is strongly diversified and resilient against significant pest and disease outbreaks. Due to this diversity, most pest/disease impacts on Darkwoods will be dispersed endemic levels with a low property-level impact that are immaterial to onsite carbon stocks.

Pests include mountain pine beetle, Douglas-fir beetle, spruce beetle, and western hemlock looper, western spruce budworm, Douglas-fir tussock moth. The Darkwoods property has a low risk of catastrophic pest outbreaks, partially because of these types of pest and disease are typically localized endemic outbreaks targeting specific species, age, or other complex conditions, and tend to result in low overall mortality to individual or groups of trees within a stand level.

Even the worst pest outbreak in B.C. recorded history, the Mountain Pine Beetle (MPB) epidemic, had a relatively minor impact on the standing stock of timber and carbon on the property simply because there is a limited amount of suitable pine. On Darkwoods, pine makes up <6% of the forest area and has been a focus of harvest over the past decade to the point that limited pine area is at risk to MPB. Even within the pine stands, the level of mortality has been relatively low (see Table 4 in the Darkwoods 2011-2012 Monitoring Report – Less than 0.07% of Darkwoods suffered >50% stand mortality, etc.). The emissions from MPB impacted stands are included in ex-ante modelling projections, and in ex-post carbon calculations through monitoring plot and updated inventory data.

Armillaria Spp. Root rot is the most significant disease in the area, primarily in the ICH zone. Root rot tends to affect small clusters of trees in certain species and age class conditions, and is generally considered a low level source of endemic stand mortality, particular in natural stands or diversified planted stands. The BC Ministry of Forests estimates the impact of root rot on timber supply to be low beginning past year 50 in managed stand conditions (BC MOF, 2002).

Darkwoods carbon modelling includes factors to account for stand level tree mortality within growth and yield projections. The modelling takes the additional conservative step of including an additional 4% standing stock mortality every 20 years to account for various endemic natural disturbance stock losses.

A detailed discussion of disturbance regimes in this region of BC can be found in (Wong, 2003), (Cirque Resource Associates Ltd., 2007).

Rating: Insignificant.

3.1.5 Pest and Disease Likelihood

Similar to many natural forests, the Darkwoods property sees an ongoing low to very low level of endemic disturbance due to pests and disease. Due to property diversity, the likelihood of catastrophic pest and disease outbreaks is very low, as evidenced by the minimal impacts of the recent massive MPB epidemic.

Rating: < Every 10 Years

3.1.6 Pest and Disease Mitigation

NCC employs forestry staff full time for the Darkwoods property, which insures an ongoing level of monitoring for significant pest or disease on the property. Darkwoods staff undertake one or more survey flights covering the entire property annually to detect any pest, disease or other natural disturbance.

The active management plans for NCC in the carbon project scenario provide tools that can be used to prevent or manage pest or disease outbreaks on the land base. For example, timber harvesting operations have been undertaken in pine stands affected or susceptible to MPB infestation as a part of active management to reduce MPB infestation severity and duration. Ongoing forest health monitoring and proactive management actions will continue to reduce the risks of catastrophic outbreaks.

The stand level data and modelling undertaken for the baseline and project scenarios includes typical operational adjustment factors to reflect average stand level tree losses due to pests, disease, and natural mortality on an annual basis. In addition, an additional 4% stand loss every 20 years has been included to conservatively reflect additional endemic natural disturbances, including pests and disease. Stands affected by MPB have been directly monitored and updated in new inventory data, which have then been isolated in specific Analysis Units for stand and carbon modelling.

Rating: Both of the Above

c) Extreme Weather:

3.1.7 Extreme Weather Significance

Although extreme weather in the form of wind, heavy precipitation and flooding, and drought do occur on Darkwoods, the diversity of the terrain and forest means impacts are localized and generally immaterial to carbon stocks. Darkwoods staff undertake one or more property survey flights to inspect for blowdown on an annual or more frequent basis. Wind damage and blowdown is isolated and insignificant – no material continuous blowdown areas have occurred on Darkwoods in the past 40 years.

The diverse forest types and ages and high level of natural stand conditions provide substantial property level resilience to wind and other extreme weather events. Extreme precipitation events do occur on Darkwoods, however the mountainous and diversified terrain means flooding is limited to localized streambeds, and generally do not materially affect surrounding forest biomass. Localized small landslides do occasionally occur on steep ground on Darkwoods, often near recent road disturbances, however these are generally very small occurrences (i.e. <0.25 ha) and rare with well planned and built road and crossings. NCC has an active road maintenance and reclamation program to reduce road disturbance risks. Drought is a potential threat on a stand or regional basis; and although this property does sit within an interior wet belt, it is possible droughts could extend the dry zones into the wet transition zone on this property. The diversity of tree species provides a long-term resilience to drought, particularly in robust natural stands. Very old stand types found across Darkwoods are evidence of a very low level of drought impact history across the property over very long timeframes. Overall, the property has no history of material levels of forest or carbon stock impacts from any type of extreme weather events.

Rating: Insignificant.

3.1.8 Extreme Weather Likelihood

Extreme weather events likely occur on Darkwoods on an annual or semi-annual basis. However, weather based damage to trees and stands are very infrequent and the level of impact on carbon stocks is immaterial.

No recordable extreme weather events affecting carbon stocks have occurred since the project inception in 2008.

Rating: < Every 10 Years

3.1.9 Extreme Weather Mitigation

NCC employs full time staff for the Darkwoods property who actively monitoring the property on an ongoing basis, and specifically after severe weather events as necessary. NCC implement mitigating management as needed to mitigate any damages, as feasible and necessary.

Any material forest impacts from weather events are recorded and updated to the forest inventory GIS system, which then is accounted for in ex-ante and ex-post carbon stock accounting as necessary. As noted, the FORECAST carbon modelling conservatively accounts for an additional 4% natural disturbance loss every 20 years in addition to typical stand level growth and yield mortality.

Rating: Both of the Above

3.1.10 Other Natural Risks

d) Geological Risk:

Not applicable - Darkwoods is not subject to any nearby geological risks from earthquake, volcano, etc.

e) Other Natural Risks

Not applicable - Darkwoods is not subject to any other material natural risks to carbon stocks.

REFERENCES:

B.C. Ministry of Forests. (2002). Kootenay Lake TSA - Rationale for AAC Determination. Nelson, BC: BC Ministry of Forests.

B.C. Ministry of Forests. (2003). Biodiversity Guidebook. Victoria, BC: Ministry of Forests - Government of British Columbia <<http://www.for.gov.bc.ca/>>

Wong, C. B. (2003). Estimating historical variability of natural disturbances in British Columbia. Victoria, BC: B.C. Min. For., Res. Br., B.C. Min. Sustain. Resource. Manage., Resource Plan. Br., Victoria, B.C. Land Manage. Handb. No. 53.